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SECTION : 5D

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Use-Case Flow specification :

Academic Elective Bidding and allocation system

Domain : campus and academic operations

Actors : student, academic registrar

Use case : run allocation algorithm

Pre-conditions :

- The elective bidding window closed for the current round.
- All student bids (ranked preferences) to be submitted and recorded.
- Seat capacity for each elective course has been set by the academic registrar.

Post-conditions :

- Every submitted bid has a final status (Allotted, waitlisted or rejected).
- Allocation results are recorded in the system and visible to each student for their own bids.
- A notification has been queued or sent to every affected student.

Ideal scenario expected :

1. Academic registrar selects “Run allocation” for the current bidding round. System displays the round summary (course list, total bids recieved, seat capacities).
2. Registrar confirms the run. System locks further bid submission for this round.
3. System validates prerequisites eligibility for every bid (<include> validate prerequisites).
4. System ranks eligible bids for each course by student priority (bid rank, then tiebreaker rule).

5. System assigns seats to the highest ranked eligible student up to each course's capacity.
6. System places remaining eligible student on a waitlist, ordered by rank.
7. System generates the final allocation result list and stores it against each record.
8. System notifies each student of their result (Allocates, waitlisted, or rejected) via portal and email.
9. System displays a completion summary to the registrar (seats filled, waitlisted count, rejected count)
10. Use case ends successfully.

If Bid rank tie between students :

- System detects two or more students with identical bid rank for the same course and remaining seat.
- System applies the configured tie breaker rule (e.g. higher CGPA, then earlier submission timestamp).
- System resolves the tie and resumes the main flow at step 5

If pre-requisites are not met :

- System finds that a student does not meet the prerequisites for a bid choice.
- System marks that specific bid choice as rejected and considers the student's next ranked preference.
- Use case resumes the main flow at step 3 for the remaining bed choices.